

Anticipatory Behavioural Responses to Environmental Policy Announcements: Evidence from the London Ultra Low Emission Zone

Candidate 68508

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Abstract

ABSTRACT

1 Introduction

Standard models of regulatory compliance assume behavioural change occurs at the point of implementation. Yet if economic agents are forward-looking and form rational expectations, as Muth (1961) first formalised, credible future policy may alter present behaviour well before implementation begins. Kydland and Prescott (1977) demonstrated that the effectiveness of policy depends fundamentally on its credibility: agents who believe a policy will be implemented adjust accordingly, while announcements perceived as reversible or discretionary will be ignored. Lucas (1976) showed more generally that behavioural models are not invariant to policy regime, since the decision rules being estimated already embed agents' expectations about future policy. Applied to the gap between announcement and enforcement, this implies that treating behaviour as static until formal implementation misspecifies the underlying decision problem. Credible policy announcements may therefore constitute a policy instrument in their own right.

The London Ultra Low Emission Zone (ULEZ) provides a natural empirical setting for studying anticipatory compliance. Introduced in April 2019 under Mayor Sadiq Khan, the scheme charges non-compliant vehicles a daily fee for entering the zone, with the stated aim of reducing the air pollution damaging residents' health and the environment. Two features make it a useful case. First, each expansion was announced well before it took effect. The London-wide expansion of August 2023, for instance, was proposed in March 2022 and confirmed that November, leaving a gap of roughly nine months between the November confirmation and enforcement. That gap provides a window in which anticipatory compliance, if it occurs, can be detected. Second, the zone expanded three times between 2019 and 2023, each announced in advance of implementation. Repeated expansions allow the anticipatory response to be tested across successive waves, and they raise a further question: whether the effect weakens as each new announcement conveys less genuinely new information than the last.

The paper addresses two questions. First, did each ULEZ expansion generate anticipatory compliance before implementation? Second, does the anticipatory effect vary across the three expansions? Both follow from the theoretical claim at the centre of this study: that a credible announcement functions as a policy instrument in its own right, capable of shifting behaviour before enforcement. If this holds, compliance should rise measurably in the window between a credible announcement and its implementation. Whether it does, and whether the response weakens as successive expansions convey less new information, is what the analysis sets out to test.

The analysis uses synthetic control as its primary method. For each treated borough, the method builds a counterfactual from a weighted combination of untreated units, which avoids reliance on the parallel trends assumption required by conventional difference-in-differences. The donor pool for each expansion draws on units not yet treated in that wave: for the 2019 and 2021 expansions this includes outer London boroughs alongside the surrounding districts, while for the 2023 London-wide expansion, where every London borough is eventually treated, the donor pool rests on the surrounding never-treated districts alone. The dataset covers 52 units: 33 London boroughs and 19 surrounding districts serving as never-treated controls.

These form a balanced quarterly panel running from 2011 Q4 to 2024 Q4. The outcome is the ultra low emission vehicle (ULEV) share of licensed cars, constructed from DVLA vehicle licensing records. The synthetic control results are tested against two-way fixed effects difference-in-differences, an event study, and interrupted time series specifications, with a geographic regression discontinuity design considered but limited by the small number of boroughs near the zone boundary.

The analysis finds anticipatory compliance in specific boroughs rather than spread evenly across London, and the effects are consistently modest. For the 2019 expansion, Tower Hamlets shows the strongest response, with its ULEV share rising above its synthetic counterfactual in the quarters after the announcement, though the effect is marginal rather than conventionally significant. The 2021 expansion yields no significant effects, with only Hammersmith and Fulham reaching the suggestive range. For the 2023 London-wide expansion, two outer boroughs show significant anticipatory effects once surrounding never-treated districts are used to build the counterfactual: Hillingdon and Sutton. Disaggregating by keepership points to a mechanism. The response comes mainly from company fleets rather than private owners, which fits the idea that fleet operators renew vehicles ahead of a known compliance deadline.

This study makes three contributions. Theoretically, it extends the literature on anticipatory behaviour, developed largely outside environmental regulation, to a domain where compliance requires a costly and durable purchase. Empirically, it tests this mechanism across three repeated waves of the same policy, where existing work has largely examined single policy events, tracing the anticipatory response as the policy moves from novel to familiar. Methodologically, it uses districts bordering Greater London as never-treated controls, a donor pool that shares the commuter exposure of outer London boroughs without being subject to the zone itself, providing a cleaner counterfactual than inner London boroughs that were themselves treated in earlier waves.

ROADMAP

2 Policy Background

The Ultra Low Emission Zone (ULEZ) is a charging scheme run by Transport for London (TfL) under the Mayor of London. Vehicles that do not meet specified exhaust emission standards pay a daily fee of £12.50 to drive within the zone, applied at all hours. Petrol cars and vans must meet the Euro 4 standard, diesel cars and vans the Euro 6 standard, and motorcycles and mopeds the Euro 3 standard. Compliance is defined by these emission standards rather than by fuel type: a Euro 4 petrol or Euro 6 diesel car is exempt from the charge, as are fully electric and most plug-in hybrid vehicles. Ultra low emission vehicle adoption is therefore one route to compliance among several, but a distinctive one, since it represents the durable, forward-looking capital decision that anticipatory behaviour would predict. The policy aims to improve air quality by lowering concentrations of harmful pollutants, particularly nitrogen dioxide and particulate matter, and to protect public health across London.

ULEZ emerged as a gradual evolution of London’s clean air policy, building on the 2003 Congestion Charge and the 2008 Low Emission Zone that targeted heavy vehicles. First

proposed in 2013 by Mayor Boris Johnson to address illegal levels of air pollution, it was accelerated and launched in central London by Mayor Sadiq Khan in 2019, driven by rising public health concerns and series of legal defeats over the UK's failure to meet air quality obligations. These earlier schemes meant London drivers were already familiar with zone-based charging, and the enforcement infrastructure of cameras and Automatic Number Plate Recognition (ANPR) was already in place. That existing apparatus made subsequent ULEZ announcement credible: a commitment to expand the zone was believable precisely because the equipment to enforce already existed.

Launched on 8 April 2019, London's initial 24-hour ULEZ was mapped to the central Congestion Charge Zone, covering roughly 21 square kilometres and encompassing the high-density commercial core. The scheme was first proposed in 2015 under Mayor Boris Johnson, but the decisive signal came on 4 April 2017, when Mayor Sadiq Khan announced the April 2019 launch date and brought the timetable forward. The same 2017 announcement also flagged a planned extension to the North and South Circular roads, which was formally confirmed on 8 June 2018 with an implementation date of 25 October 2021. This April 2017 announcement is the point from which anticipatory responses to the central zone are measured in the analysis that follows.

The inner expansion to the North and South Circular roads was confirmed in June 2018 and implemented on 25 October 2021. Charges and emission standards remained identical to the central zone, but the scale increased sharply, covering an area roughly eighteen times larger and bringing approximately 3.8 million residents within the zone [Mayor of London Press release]. The gap of over three years between confirmation and enforcement is the longest of the three expansions, giving the widest window in which anticipatory compliance could occur.

The scheme was then proposed for a London-wide expansion in March 2022, confirmed in November 2022, and implemented on 29 August 2023, leaving a nine-month window between confirmation and enforcement. This was the largest expansion, bringing all 32 boroughs and the City of London within the zone. Its credibility was tested directly: a coalition of five councils (four outer London boroughs - Bexley, Bromley, Harrow, and Hillingdon, and Surrey County Council) challenged the expansion in the High Court, but the challenge failed in July 2023 and enforcement proceeded as announced. That the commitment held despite legal challenge reinforced the signal to drivers that the deadline was firm.

The credibility of these commitments rested on three reinforcing foundations. Each expansion was preceded by statutory public consultation and formal confirmation, signalling that the policy followed a legally prescribed procedure rather than an arbitrary mayoral decision. The commitment to charge non-compliant vehicles was underpinned by the camera and ANPR infrastructure already in place, so the announced restrictions could be monitored and penalties imposed from the first day of enforcement. Most decisive was demonstrated follow-through: the 2019 and 2021 expansions were both delivered on their announced timetables, so by the time the 2023 expansion was confirmed, drivers had twice seen the Mayor commit to a future extension and carry it out. Announcements that might once have been read as provisional intentions had become commitments road users could expect to be honoured. Credible, pre-announced expansions created precisely the conditions under which forward-looking individuals would adjust their behaviour before enforcement began.

3 Literature Review

This literature review establishes the foundations for the dissertation in three stages. It first develops the theoretical framework, which explains why credible policy announcements should induce anticipatory behavioural responses before implementation. It then reviews the existing empirical evidence on anticipatory behaviour and low-emission zone policies, assessing what is already known and where important questions remain unanswered. Finally, it turns to the methodological literature, outlining the empirical strategies capable of identifying such anticipation and motivating the research design adopted here. The review concludes by identifying the gap at the intersection of these literatures that this paper seeks to address.

3.1 Theoretical Framework

The theoretical foundation for anticipatory behavioural responses begins with the premise that economic agents make decisions using information about both current and expected future conditions. John F. Muth’s rational expectations hypothesis provides the micro-foundation for this argument. Muth (1961) proposed that agents’ expectations of future economic variables are not systematically biased but instead coincide, on average, with the predictions implied by the relevant economic model, conditional on the information available to them. Expectations formed this way efficiently incorporate all available information, including credible signals about future policy, rather than relying solely on past experience or adjusting only once change occurs. Although Muth developed this framework in the context of commodity price expectations, its logic extends naturally to announced regulatory costs. When policymakers communicate a future increase in the cost of owning or operating a non-compliant vehicle, that information becomes relevant immediately rather than only when the regulation enters into force. A forward-looking vehicle owner therefore has an incentive to evaluate the future costs of retaining a non-compliant vehicle against the costs of replacing or scrapping it before enforcement begins. Under rational expectations, disregarding information already revealed would itself be irrational, even though the timing of any individual response depends on the costs of adjustment.

This forward-looking behaviour has important implications for how policy effects should be understood empirically. If agents incorporate anticipated policy into present decisions, then observed behaviour before implementation may already reflect the future policy environment. Lucas (1976) formalised this insight through the Lucas critique, arguing that the estimated relationships used to evaluate policy cannot be assumed to remain invariant across policy regimes, because those relationships already embed agents’ expectations about the policy environment. Once a future policy is announced, agents revise their decision rules to reflect the expected future environment. Consequently, the period preceding implementation is not necessarily an undisturbed baseline against which policy effects can simply be measured. While the Lucas critique is often invoked as a warning against evaluating policy using naive before-and-after comparisons, it is more constructive in the present context. The possibility that behaviour changes before enforcement is not a methodological nuisance to be eliminated but the phenomenon of substantive interest. Rather than treating pre-implementation adjustment as a source of bias, this paper asks whether the announcement induced measurable changes in vehicle compliance before the ULEZ expansion came into force.

Whether such anticipatory adjustment occurs, however, depends on a further condition. Rational agents will respond to announced future policy only if they believe that the policy will in fact be implemented as announced. Kydland and Prescott (1977) demonstrate that expectations depend not only on policymakers' announcements but also on the credibility of those announcements. If agents expect a policy to be delayed, reversed, or weakened before implementation, the future regulatory cost remains uncertain and there is little reason to incur the costs of adjustment in advance. Conversely, where a policy commitment is regarded as credible and durable, the expected future cost becomes sufficiently certain to influence present behaviour. Credibility therefore provides the mechanism linking announcements to anticipatory responses. This insight is particularly important for understanding differences across successive ULEZ expansions. Variation in anticipatory compliance across waves need not imply that rational expectations fail; rather, it may reflect differences in the informational content or perceived credibility of successive announcements. As credible commitments accumulate over time, later announcements may convey less genuinely new information, reducing the scope of additional anticipatory adjustment even when the policy itself remains fully credible.

Taken together, these three contributions generate a coherent theoretical framework for analysing anticipatory behavioural responses. Rational agents incorporate credible information about future policy into current decision-making; those expectations imply that behavioural adjustment may begin before implementation, meaning that the pre-implementation period cannot automatically be treated as an unaffected baseline; and such adjustment occurs only when policy announcements are regarded as credible. The resulting prediction is therefore conditional rather than universal: where a credible announcement conveys precise information about a future regulatory cost, forward-looking agents have an incentive to adjust before implementation when the expected gains from acting early outweigh the costs of adjustment. Anticipatory responses should therefore be expected to be partial and heterogeneous rather than uniform across all individuals or policy settings. Although the components of this theoretical chain have been examined separately across a range of economic contexts, they have not been brought together to study anticipatory responses to geographically bounded environmental regulation in the window before implementation — the intersection this dissertation addresses.

4 Data and Study Design

5 Results

6 Robustness Checks

7 Heterogeneity Analysis

8 Discussion and Conclusion